

Project Overview — Evaluation, Metrics, and Risks of Hybrid Datasets

Research question: Given a known inference distribution, how well can pre-training distribution-shift metrics between a hybrid training dataset and a real inference dataset predict downstream model performance?

Hypothesis: the stronger the measured distribution shift between training and inference data, the weaker the model's performance at inference time.

Why this matters: naively testing every possible real/synthetic training mix by training a model on each and checking inference performance is expensive. If distribution-shift metrics, measured *before* training, reliably predict that performance, the best hybrid composition could be preselected without training every option.

The three datasets

Dataset	Role	Description
Cityscapes	Real / inference baseline	Real urban street scenes, 19-class semantic segmentation. Both the source of "real" training data and the fixed inference (test) set.
Synscapes	Synthetic — smaller reality gap	Photorealistic, physically-based rendering, explicitly designed to match Cityscapes' distribution.
GTA5	Synthetic — larger reality gap	Game-engine graphics, American street layouts — visually and semantically further from Cityscapes than Synscapes.

All three share the same 19-class label set, are publicly available, and come from the same target domain (urban street scenes), which is why they were selected as the dataset triple for this study.

What "hybrid dataset" means here

Ten total training-dataset configurations are constructed: five real/synthetic compositions × two synthetic datasets (Synscapes, GTA5). Total training-set size is held constant across all configurations so that performance differences are attributable to distribution shift, not dataset size.

Configuration	Composition	Expected shift from Cityscapes (inference)
Real baseline	100% real	Lowest
Hybrid 1	75% real / 25% synthetic	Low
Hybrid 2	50% real / 50% synthetic	Medium
Hybrid 3	25% real / 75% synthetic	High
Synthetic baseline	100% synthetic	Highest

This 5-configuration ladder is built once with Synscapes and once with GTA5 → **10 configurations total.**

Pre-training metrics (measuring shift before training)

Metric	Backbone	Assumes Gaussian?	Focus
FID (Fréchet Inception Distance)	Inception-V3	Yes	Style
KID (Kernel Inception Distance)	Inception-V3	No (kernel-based)	Style — better estimator than FID at small sample sizes
CLIP embedding distance	CLIP (e.g. ViT-B/32)	No (centroid-only)	Semantic — compares only cloud centers
FCD (Fréchet CLIP Distance)	CLIP	Yes	Semantic — also considers distribution shape

Raw scores are **not comparable across metrics** (different scales, no fixed range). Interpretation focuses on ranking and correlation with downstream performance within each metric, compared against the real/synthetic baselines — not on comparing absolute values between metric families.

Audience-suggested extensions (not yet core to the design):

- Lower-level, embedding-free metrics — e.g. pixels-per-class annotation statistics, to capture class-distribution shift directly from labels.
- Domain-specific embedding backbones (e.g. DINOv2, StreetCLIP) as Inception-V3 replacements in FID/KID, since Inception-V3 was never trained on urban/street imagery.

Evaluation approach (end-to-end)

1. **Construct** the 10 hybrid training datasets (fixed total size, controlled injected shift).
2. **Quantify** distribution shift for each configuration using each selected pre-training metric.
3. **Train** a segmentation model — fixed architecture, fixed training procedure — separately on each of the 10 configurations, then evaluate it on the fixed Cityscapes inference set (metric: presumably mIoU).
4. **Correlate** each metric's pre-training shift scores against the corresponding post-training performance, across the 10 configurations, checking for the hypothesized negative correlation (higher shift → lower performance).

Note: the segmentation-training step is **separate** from any embedding-extraction prototyping — the domain-specific embedding backbones (DINOv2/StreetCLIP/etc.) are feature extractors for the shift metrics only; they play no role in segmentation itself.

What "success" / expected results look like

- Identification of **which pre-training metrics** successfully track the known, controlled ranking of distribution shift across the 10 configurations (i.e. do the metric scores order the configurations the way the real/synthetic ratios predict?).
- Evidence for or against a **reliable negative correlation** between measured pre-training shift and downstream mIoU.
- A discussion of extending the approach to the harder, more realistic case where the inference distribution is **not** fully known in advance (this study assumes it is).
- Secondary discussion: other factors affecting hybrid-data utility, such as training scheme (combined real+synthetic training vs. synthetic-then-real sequential training).

What is *not* in scope for this study

- Reducing or mitigating distribution shift (only measuring it).
 - Varying model architecture, training hyperparameters, or training approach — these are held fixed.
 - The case of an unknown/uncertain inference distribution (acknowledged as future work).
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Timeline

Phase	Dates
Project presentation	Jun 10
Designing experiments	Jun 10 – Jun 30
Conducting experiments	Jul 1 – Jul 31
Intermediate presentation	Jul 8
Write the paper	Aug 1 – Aug 31
Final presentation	Sep 30